

# Two-Stage Machine Learning Framework for G10 FX Fair Value and Automated Trading

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# Motivation & Research Question

- Aim: to estimate fair values on FX
    - Currencies have no intrinsic value.
    - Predicting FV is substantially harder than other asset classes.
  - Then, automate a trading strategy and improve it with ML.
  - Meese and Rogoff (1983): *“a random walk model performs as well as any estimated model at one to twelve month horizons”*.
  - Question: Can supervised learning identify a stable short-term Fair Value whose deviations predict significant, tradable out-of-sample mean reversion in G10 FX?
  - Compare OLS with regularized, tree-based, stacked, and RL approaches.
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# Dataset

| Broad FX trend                                                                                   | Yield spread                                                                                                                                                                                               | Equities                                                                                                                                                                                                                                                                                          | Commodities                                                                                                                                                                                                                                                                   | Global Risk                                                                                                 |
|--------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• BBDXY</li><li>• EMFX Index</li><li>• Asia EMFX</li></ul> | <ul style="list-style-type: none"><li>• 2y yield</li><li>• 5y yield</li><li>• 10y yield</li><li>• 3m1m forward OIS</li><li>• 6m1m forward OIS</li><li>• 1y1m forward OIS</li><li>• Real 2y yield</li></ul> | <ul style="list-style-type: none"><li>• Local - S&amp;P500</li><li>• Local - Wilshire</li><li>• Local - Dow Jones</li><li>• MSCI World - S&amp;P500</li><li>• MSCI World - Wilshire</li><li>• MSCI World - Dow Jones</li><li>• MSCI World ex-US</li><li>• MSCI EM</li><li>• Local Index</li></ul> | <ul style="list-style-type: none"><li>• BCOM Index</li><li>• Gold</li><li>• Oil</li><li>• UK Natural Gas</li><li>• TTF Natural Gas</li><li>• NY Natural Gas</li><li>• Copper COMEX</li><li>• Copper LME</li><li>• Local Commodities 1</li><li>• Local Commodities 2</li></ul> | <ul style="list-style-type: none"><li>• VIX</li><li>• USTs Volatility</li><li>• G10 FX Volatility</li></ul> |



# Preprocessing

- Synthetic BBDXY: this variable was recreated for currencies in the index, e.g. BBDXY ex-EUR.
    - However, BBDXY indices were removed due to multicollinearity.
  - Forward rates were dropped for SEK, NOK, AUD, NZD, CHF, JPY.
  - ADF test: Natural Gas, Australian commodities and Global Risk variables exhibited Stationarity  $I(0)$  so were left as levels.
    - Daily log returns on variables that were non-stationary  $I(1)$
    - Some yield spreads were  $I(0)$  but all were transformed to daily value changes for consistency.
  - Forward-fill, aligned dates and created 1y rolling z-scored versions of drivers for the linear models.
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# Stage 1: Adaptive Feature Selection

- *rolling\_univariate\_ols()* performs OLS regressions to compute 250-day rolling betas and p-value significance levels for each candidate driver.
  - The diversified top-driver map keeps the strongest significant drivers across macro categories rather than allowing one theme to dominate
    - Minimum 95% significance level
    - Only one driver per category
    - 3 drivers max with highest absolute betas
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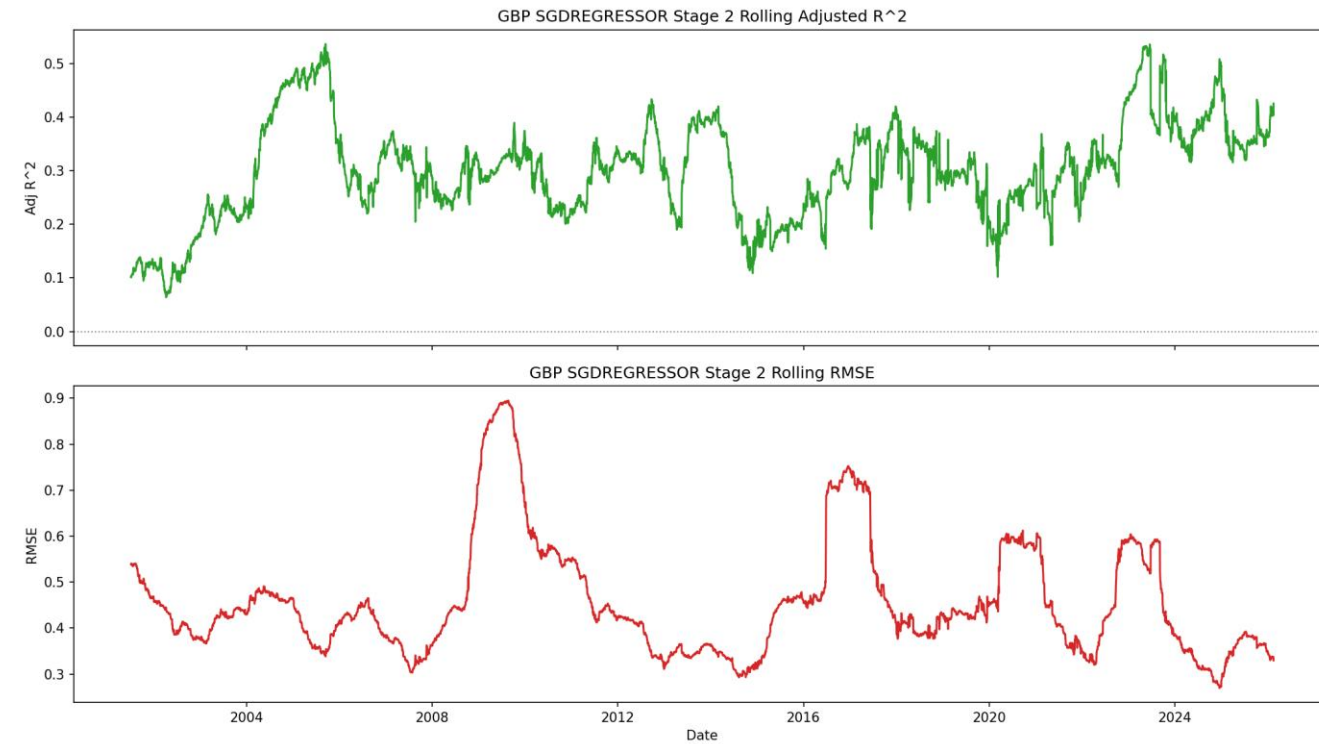
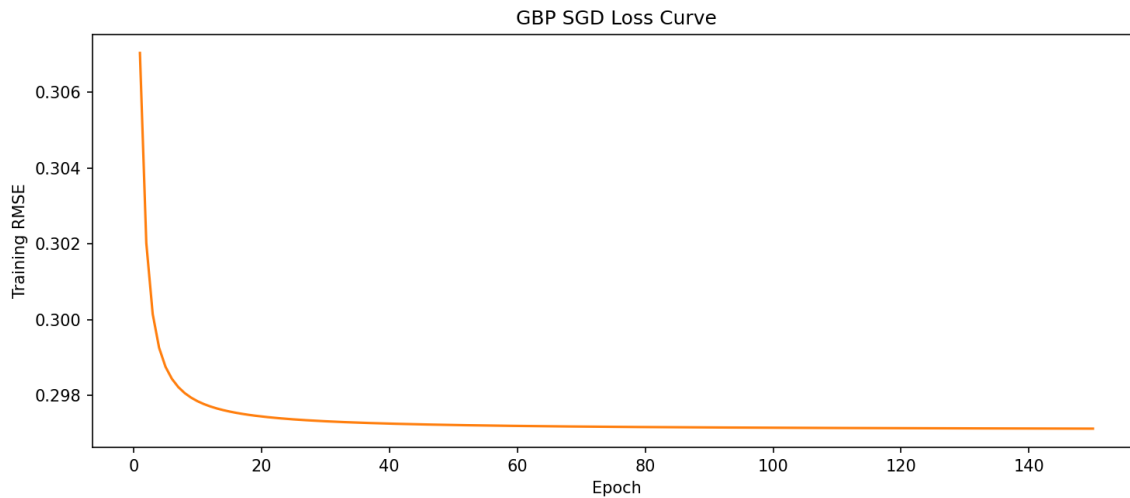
## Stage 2: Multivariate Estimations

- Walk-forward estimation suite coded in *run\_stage2\_model()*
  - Baseline: OLS
  - Regularized: RidgeCV, LassoCV, ElasticNetCV, SGDRegressor
  - Gradient Boosting: XGBRegressor, LGBMRegressor
  - Combination: Stacked ensemble.
  - Hyperparameters tuned with 4 CV folds and retuned every 60 days.
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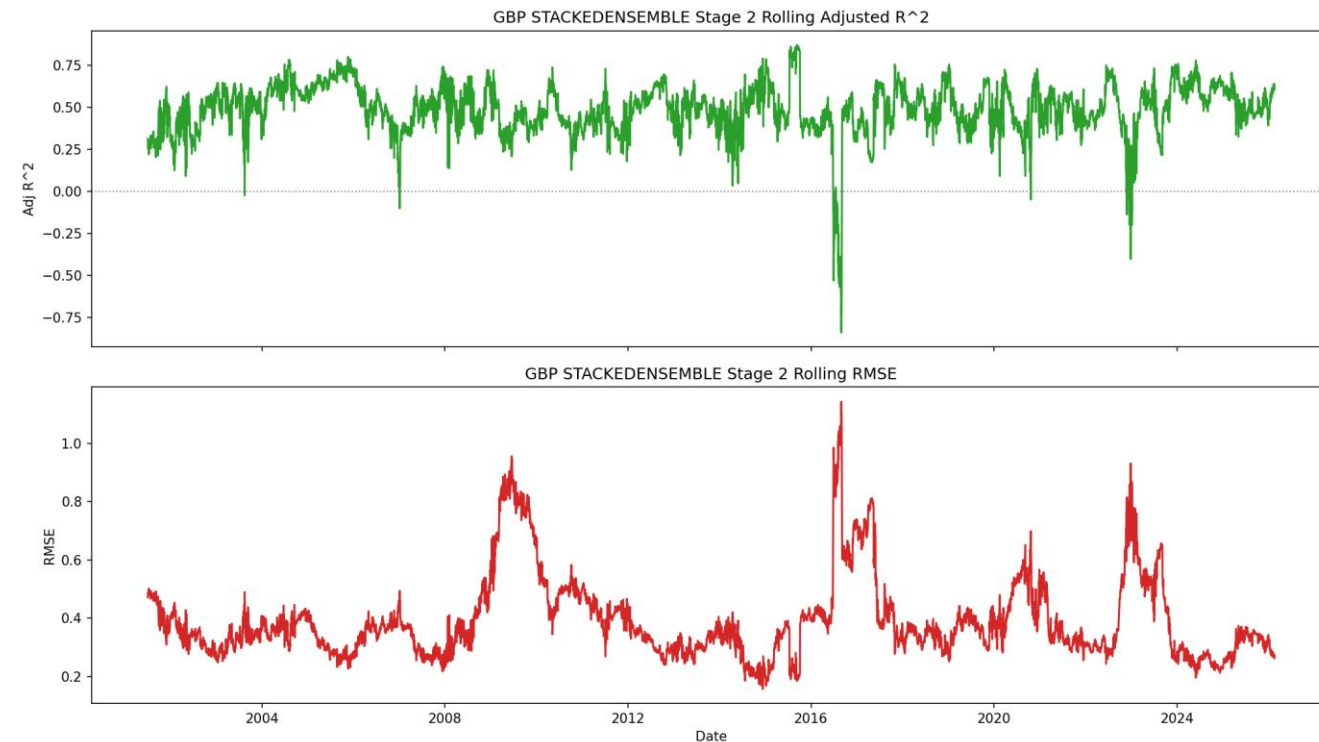
# Stochastic Gradient Descent

$$\beta^{(k+1)} = \beta^{(k)} - \eta \nabla_{\beta} \mathcal{L}(\beta^{(k)}; \mathbf{x}_{i_k}, y_{i_k})$$



# Stacked Ensemble

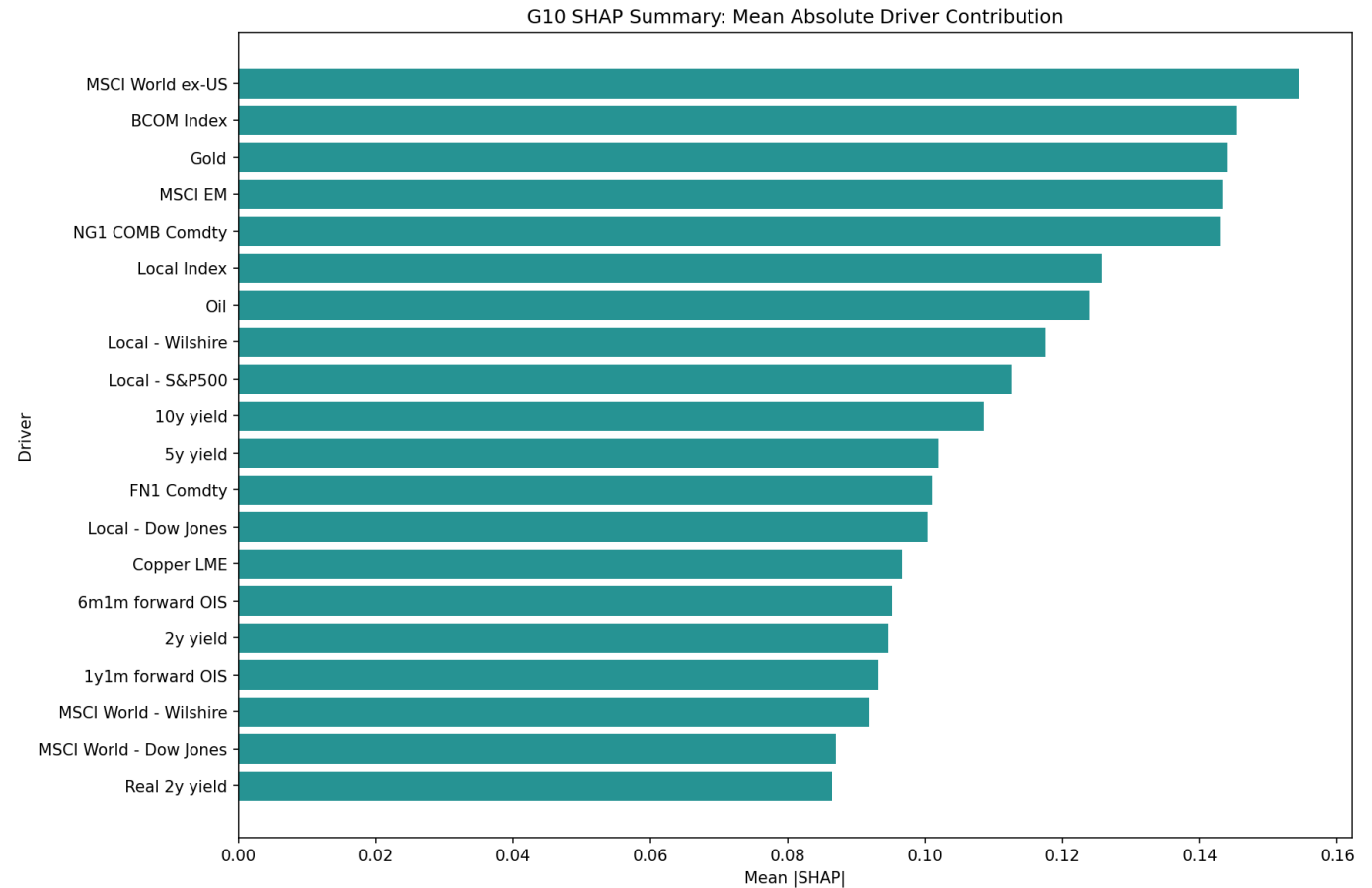
- ElasticNet: standardised data, generally low  $l1/l2$ , low  $\alpha$ .
- XGBoost and LightGBM:
  - raw data
  - `max_depth` {2, 3}
  - `n_estimators` {200, 400}
  - `learning_rate` {0.03, 0.1}
  - `subsample` {0.8, 1.0}
- First, out-of-fold predictions for each model become the meta features.
- Then, a Ridge regression fits those meta features.





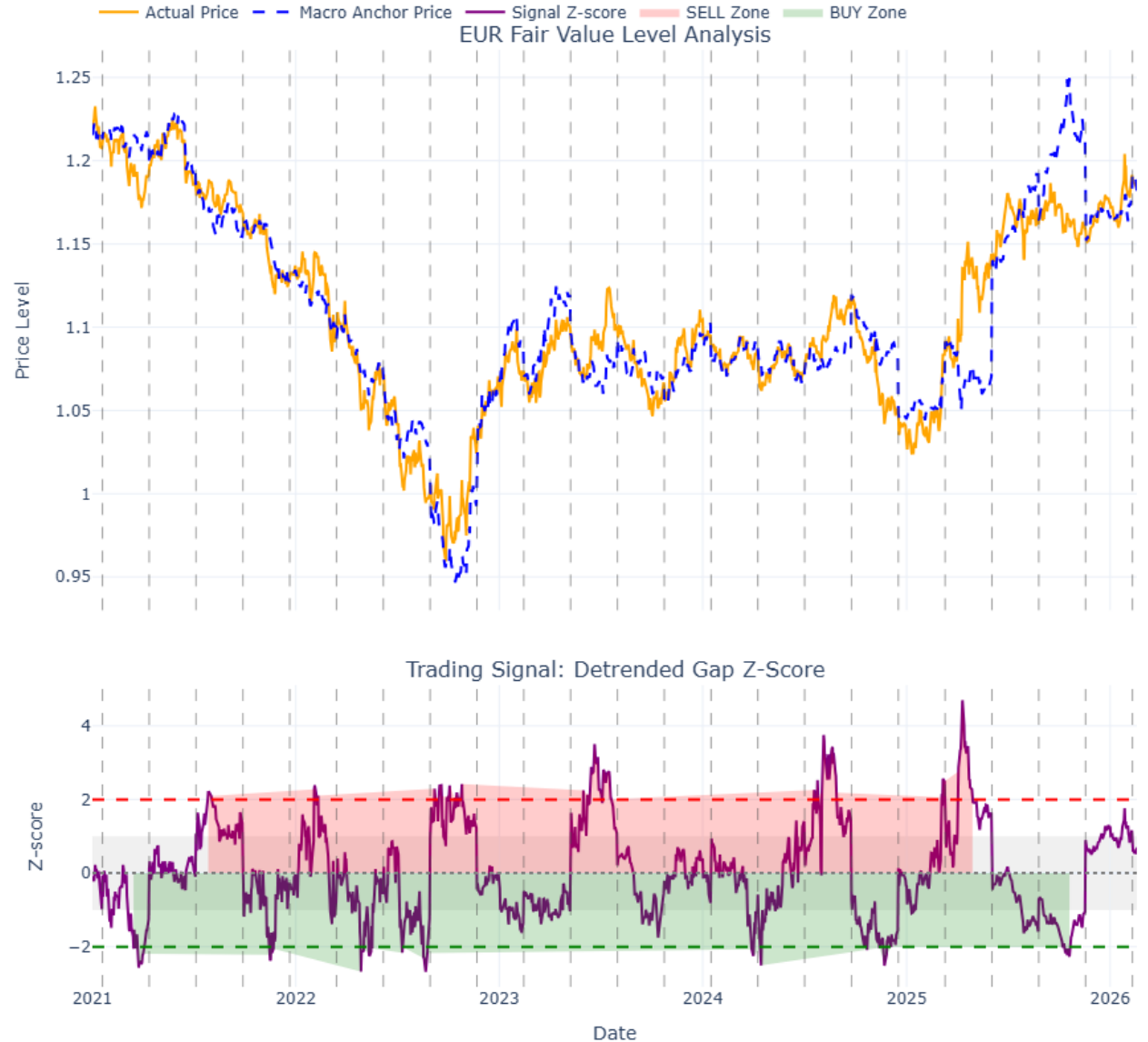
## Key drivers

- Regularized models typically favoured yield spreads.
- Tree-based models did not favour yield spreads.
- Both types had a tendency towards:
  - MSCI World
  - BCOM Index
  - Gold



# Trading Strategy

- Model determines next day's predicted return.
- These are added cumulatively to the Price Level and reset every 60 days to prevent excessive deviations.
  - Fair Value (Macro Anchor).
- Cumulative 10-day rolling sum of residuals between predicted Fair Value and Actual Price.
  - Then, standardize the residuals.
- Buy: residual z-score  $< -2$
- Sell: residual z-score  $> 2$





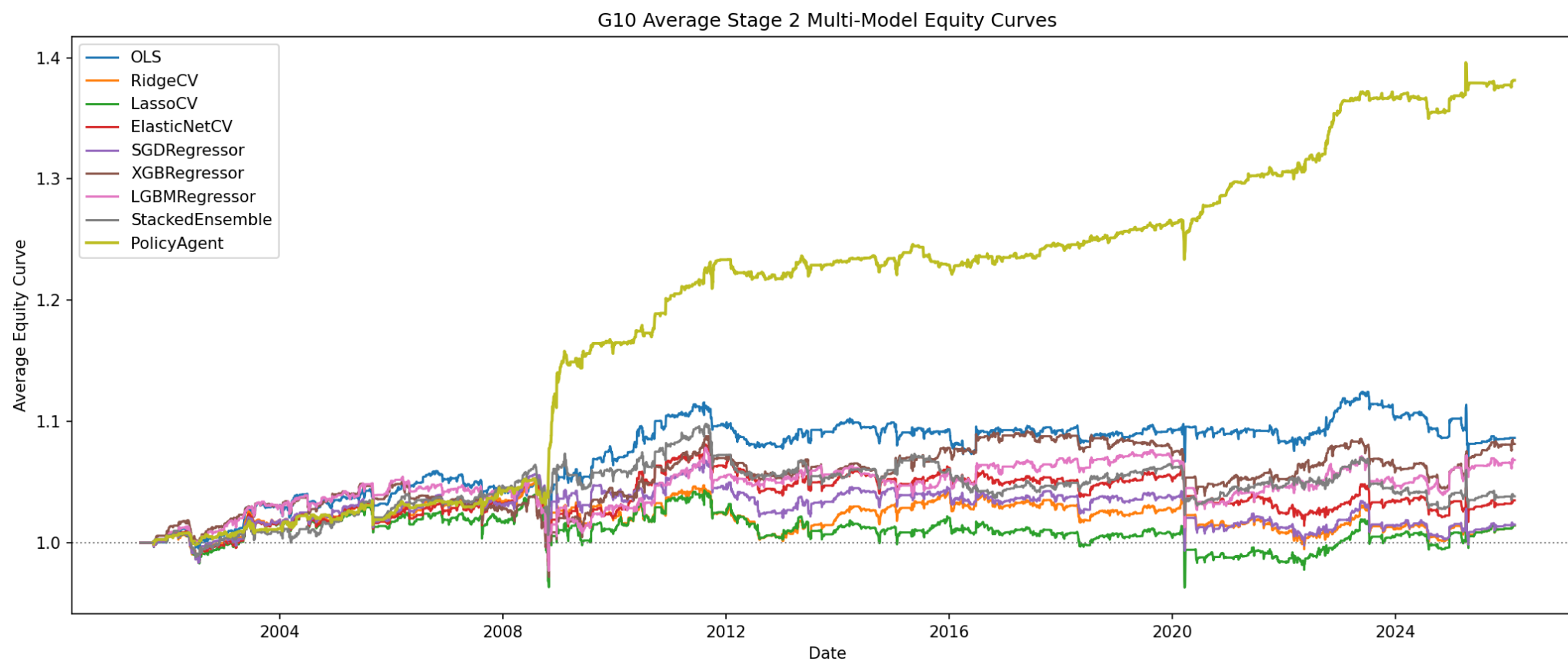
# Q-Learning Policy Agent

- Reinforcement learning was used as a policy overlay, not as the fair-value model itself.
  - The agent sits on top of the stacked ensemble and learns when to trade, not how to forecast FX returns.
  - It uses tabular Q-learning with a discrete state made from:
    - Signal\_Z = standardized valuation gap
    - Adj\_R2 = current model-fit quality
    - Volatility = recent realized FX volatility
  - Adjust the entry threshold by -0.5, 0.0, or +0.5
  - Reward is based on next-day PnL
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# Model Results

| Final_Rank | Model_Family | Model           | Average_Information_Coefficient | Average_Hit_Rate_Pct | Average_Sharpe_Ratio | Average_Maximum_Drawdown_Pct | Average_Generalization_Gap | Average_Strategy_Total_Return_Pct | Average_Adj_R2 | Average_Rolling_R MSE |
|------------|--------------|-----------------|---------------------------------|----------------------|----------------------|------------------------------|----------------------------|-----------------------------------|----------------|-----------------------|
| 3          | GBM          | XGBRegressor    | -0.012                          | 49.56                | 0.093                | -17.40                       | 0.102                      | 8.14                              | 0.43           | 0.47                  |
| 4          | GBM          | LGBMRegressor   | -0.011                          | 49.61                | 0.060                | -17.41                       | 0.086                      | 6.81                              | 0.39           | 0.48                  |
| 2          | Baseline     | OLS             | -0.003                          | 49.65                | 0.088                | -15.68                       | 0.036                      | 8.65                              | 0.34           | 0.51                  |
| 9          | Regularized  | LassoCV         | -0.001                          | 49.66                | 0.016                | -17.82                       | 0.037                      | 1.42                              | 0.32           | 0.51                  |
| 5          | Regularized  | RidgeCV         | -0.004                          | 49.32                | 0.028                | -16.83                       | 0.037                      | 1.41                              | 0.33           | 0.51                  |
| 1          | RL           | PolicyAgent     |                                 | 50.86                | 0.368                | -9.94                        |                            | 36.78                             |                |                       |
| 6          | Regularized  | ElasticNetCV    | -0.003                          | 49.34                | 0.049                | -16.94                       | 0.035                      | 3.49                              | 0.32           | 0.51                  |
| 8          | Regularized  | SGDRegressor    | -0.002                          | 49.05                | 0.025                | -18.02                       | 0.036                      | 1.49                              | 0.33           | 0.51                  |
| 7          | Stacked      | StackedEnsemble | 0.002                           | 49.06                | 0.059                | -18.59                       | 0.131                      | 3.82                              | 0.51           | 0.43                  |

# PnL Performance





# Challenges & Criticisms

- Very weak signal at short FX horizons
    - Macro fundamentals explain only a limited share of day-to-day moves.
  - Structural instability and regime shifts make fixed relationships unreliable.
  - The more complex models add tuning risk and can overfit the walk-forward pipeline.
  - The RL layer is promising, but the state space and training sample are still limited.
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# Next Steps

- Test with fully enforcing ADF and cointegration tests as a pre-filter.
  - Separate hyperparameter tuning from final testing more strictly:
    - Inner time-series CV loop for model selection.
    - Outer walk-forward window for true out-of-sample evaluation.
  - Include bid-ask prices, potentially expand to intra-day data.
  - Explore volatility regimes using a Markov switching model.
  - Expand model pipeline to G10 crosses and Emerging Market currencies.
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# Thank you!

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